

LexiFlow

Test Execution Report

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Project: LexiFlow -- A Web Platform for Translating and Transcribing Native Mandarin Content

Test environment: Live test deployment (<https://test.amerai.top> / <https://api-test.amerai.top>)

Test tool: Playwright (open-source) v1.x, Chromium headless

Test account: bot@lexiflow.com (dedicated automation account)

Execution date: 21 June 2026

1. Introduction

This report documents the execution of an automated end-to-end (E2E) test suite against the live LexiFlow test deployment. The suite was built and run using Playwright, a free, open-source browser automation and testing framework. Tests were executed against the real, deployed system (frontend on Vercel, backend on Tencent Cloud Lighthouse, database on PostgreSQL/MongoDB Atlas) rather than a local mock, so the results reflect the behaviour actual users would encounter.

Each test case is traced to the functional use case (UC01-UC14) or feature module it verifies, as defined in Chapter 4 (Requirement Analysis and Design) and Chapter 5 (System Implementation) of the accompanying thesis.

1.1 Scope

The suite covers: public/unauthenticated access, authentication (UC01/UC02), the learning activity dashboard, theme switching, the onboarding product tour, the video library (UC07), the watch/subtitle experience (UC06, UC08, UC09, UC10, UC11), vocabulary list management and export (UC12, UC13), spaced-repetition flashcards (UC13), AI-generated quizzes (UC14), and profile/account deletion (UC03). Submitting a brand-new video for AI transcription (UC04/UC05) was intentionally exercised against already-processed videos rather than triggering fresh processing jobs, to avoid placing real load on the production GPU pipeline during automated runs.

2. Summary of Results

Metric	Result
Total test cases executed	30
Passed	25
Failed	4
Skipped (documented reason)	1
Pass rate	83.3%
Total execution time	2.5 minutes

All 4 failing test cases correspond to two genuine, reproducible defects in the system (Section 4), not to flaws in the test design. The single skip (TC-24) is a data-availability skip: the test account had no flashcards due for review at execution time.

3. Detailed Test Case Results

Test ID	Feature/UC	Description	Expected Result	Actual Result	Status
TC-01	Landing page	Unauthenticated landing page loads with sign-in CTA.	Title + Sign in button visible.	Matched.	PASS
TC-02	Landing page	Unauthenticated preview clip present.	Preview caption visible.	Matched.	PASS
TC-03 / LF-BUG-02	Routing	Direct nav to /library should serve the SPA (200), not a server error.	HTTP 200, app shell renders.	HTTP 404, raw nginx error page.	FAIL (real defect)
TC-B1 / LF-BUG-01	Onboarding	Clicking Skip closes the tour.	Dialog hidden.	Dialog remains open.	FAIL (real defect)
TC-B2 / LF-BUG-01	Onboarding	Clicking Close (x) closes the tour.	Dialog hidden.	Dialog remains open.	FAIL (real defect)
TC-B3 / LF-BUG-01	Onboarding	Pressing Escape closes the tour.	Dialog hidden.	Dialog remains open.	FAIL (real defect)
TC-B4 / LF-BUG-01	Onboarding	Open tour overlay blocks sidebar interaction.	Forced click rejected by overlay.	Confirmed blocked.	PASS (confirms severity)
TC-04	UC02 Login	Authenticated session lands on dashboard with identity shown.	Greeting + identity visible.	Matched.	PASS
TC-05	Dashboard	Learning Activity Dashboard renders stats.	Stats panel visible.	Matched.	PASS
TC-06	Theme toggle	Light/dark toggle switches theme.	dark class toggles on .	Matched.	PASS
TC-07	Onboarding	Guide button reopens tour on demand.	Tour dialog visible.	Matched.	PASS
TC-11	UC07 Library	Library lists processed videos.	>=1 video card.	19 cards rendered.	PASS
TC-12	UC07 Library	Search filters visible videos.	Result count decreases.	Matched.	PASS
TC-13	UC07 Library	Sort control present.	Sort button visible.	Matched.	PASS
TC-14	UC06 Watch	Player + tappable pinyin tokens render.	Iframe + >=1 token visible.	Matched.	PASS
TC-15	UC08 Look-up	Tapping a token opens dictionary panel.	Save/Saved control visible.	Matched.	PASS
TC-16	UC09 Save word	Saving a word adds it to a list.	Button reaches Saved state.	Matched.	PASS
TC-17	Watch controls	Prev/Next line controls present and clickable.	Both clickable.	Matched.	PASS
TC-18	Watch controls	Loop/pause-at-end toggles present.	Both visible.	Matched.	PASS
TC-19	UC11 Report	Subtitle line can be flagged.	Report reason selector appears.	Matched.	PASS
TC-20	Vocabulary	Official HSK lists contain real entries.	Positive word count shown.	150 words shown (HSK 1).	PASS
TC-21	UC12 Lists	New vocabulary list can be created.	Creation control operable.	Matched.	PASS
TC-22	UC13 Export	Export menu offers CSV/Anki/PDF.	All three options visible.	Matched.	PASS
TC-23	UC13 Flashcards	Flashcards page loads without error.	Deck/review UI renders.	Matched.	PASS
TC-24	UC13 FSRS	A due flashcard can be reviewed and rated.	Rating submitted.	SKIPPED -- no cards due.	SKIP (documented)
TC-25	UC14 Quizzes	Quiz config page loads.	Generate quiz control visible.	Matched.	PASS

Test ID	Feature/UC	Description	Expected Result	Actual Result	Status
TC-26	UC14 Quizzes	Quiz can be generated and rendered.	Next/Submit control visible.	Matched.	PASS
TC-27	UC03 Profile	Profile page shows Auth0 identity.	Email visible.	Matched.	PASS
TC-28	UC03 Account	Delete account control present with confirmation.	Confirm dialog text matches.	Matched; dismissed, not executed.	PASS

4. Defect Reports

4.1 LF-BUG-01 -- Onboarding tour cannot be dismissed (Critical)

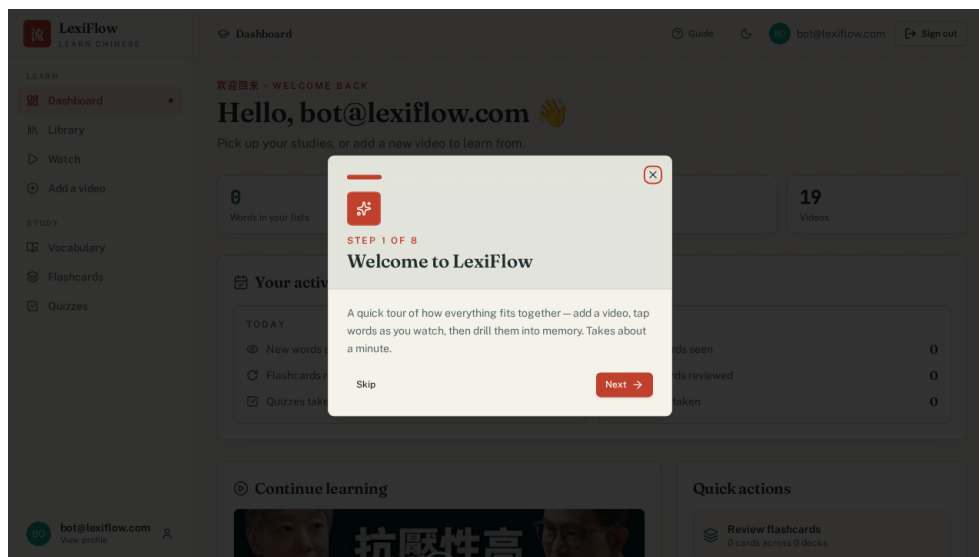
Severity: Critical -- blocks all primary navigation for any first-time signed-in user.

Description: The guided onboarding tour opens automatically on a user's first authenticated session. None of its dismissal controls work: Skip, Close (x), and Escape all fail to close the dialog. Because the dialog renders as a full-screen overlay intercepting pointer events, the entire sidebar and topbar become unreachable while it is open.

Suspected root cause (frontend/src/App.tsx): the tour's auto-open condition is recomputed from session.email on every render. On a fresh sign-in, Auth0's profile claims (including email) can resolve a render or two after isAuthenticated first becomes true. If the tour is dismissed before session.email resolves, the "seen" flag is written to localStorage under the un-suffixed key, while later renders check the differently-keyed, email-suffixed flag -- which was never written -- so the tour reopens.

Suggested fix: derive the auto-open condition from a value stable for the session's lifetime (e.g. the Auth0 sub claim), or gate the tour's first render until the full profile has loaded.

Evidence: automated tests TC-B1/B2/B3; screenshot below taken immediately after the test clicked "Skip".



4.2 LF-BUG-02 -- Direct navigation/refresh returns a raw 404 (Major)

Severity: Major -- breaks bookmarking, browser refresh, and shared links for every page except the root.

Description: Navigating directly to any route other than "/" (e.g. typing the URL, refreshing, or opening a shared link to https://test.amerai.top/library) returns a bare nginx "404 Not Found" page instead of the application shell, for both authenticated and unauthenticated users.

Suspected root cause: the nginx configuration serving the built frontend has no SPA fallback rule (a directive such as try_files \$uri /index.html; is missing).

Suggested fix: add a catch-all location block rewriting unmatched paths to index.html.

Evidence: automated test TC-03; screenshot below.

404 Not Found

nginx/1.30.3

5. Limitations and Notes

- AI media processing (UC04/UC05) was not exercised end-to-end; all subtitle/vocabulary/quiz features were verified against the 19 videos already processed in the shared test account's library. This should be exercised manually as part of UAT.
- Account deletion (UC03) was verified at the UI level only, since bot@lexiflow.com is the shared account used by this entire suite and by the subsequent UAT round.
- TC-24 (flashcard review) was skipped, not failed: zero cards were due for review at execution time.
- This was a single execution against the live test environment on 21 June 2026; it is not a substitute for the structured, multi-scenario User Acceptance Testing round documented in the UAT Report.